L-T-776a
May 31, 1966

SUPERSEDING
Int. Fed. Spec. L-T-00776 (GSA-FSS)
June 15, 1964

FEDERAL SPECIFICATION

TRAY, SERVICE (LAMINATED FIBER GLASS, POLYESTER RESIN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers polyester resin fiber glass reinforced serving trays. They are not intended for direct contact with food for human consumption.

1.2 Classification. Trays shall be of the following types and sizes, as specified (see 6.2):

Type I—(Rectangular).
Size 1—20-1/2 by 15-1/4 inches.
Size 2—18 by 14-1/16 inches.

Type II—(Trapezoidal).
Size 1—18 by 14 inches.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-553—Paper, Wrapping, Tissue.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-640—Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Federal Food and Drug Administration Publication:

Federal Food, Drug, and Cosmetic Act and General Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS.

3.1 Preproduction sample. When specified (see 6.2), four trays manufactured from materials and in the facilities to be used for production shall be submitted or made available to the contracting officer or his authorized representative for the inspection specified in 4.1.1. The approval of the preproduction sample authorizes the start of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. Upon approval, the sample trays shall be used as standard samples for determining conformance with color, surface finish, and gloss requirements (see 3.13, 3.14, and 3.15).

3.2 Material. Where material is not covered by a specification, it shall be of the quality normally used in commercial practice. All materials shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and General Regulations Promulgated Thereunder.

3.2.1 *Glass fibers.* Glass fibers shall be new, first quality electrical glass, type E. The glass content of the tray shall be 35 ± 5 percent by weight (see 6.5).

3.2.1.1 *Veil.* The top and bottom of the serving trays shall have an overlay of nonwoven colorless continuous filament glass fiber veil. An additional veil of colored continuous filament glass fiber may be placed over the core under the top and bottom veil at the option of the manufacturer.

3.2.1.2 *Core.* The core shall consist of glass fibers 1/2 to 2 inches in length either in mats of two to four flat layers for forming or one preformed shape. The weight of the core shall be 3 to 4.5 ounces per sq. ft.

3.2.2 *Reinforcement wire.* At the option of the manufacturer, reinforcement wire shall be aluminum, corrosion-resistant steel, or carbon steel electrodeposited with a corrosion-resistant metal. Diameter of the wire shall be 0.165 ± 0.010 inch for aluminum and 0.148 ± 0.010 inch for steel.

3.2.3 *Polyester resin.* Polyester resin shall be virgin materials of unsaturated polyester alkyd, modified with not more than 30 percent monomeric styrene. The filler or pigment content shall be no greater than one percent of the total weight of resin used.

3.2.4 *Bacteriostatic retardance.* Trays shall contain sufficient chemical agents to insure a bacteriostatic retardance to the growth of bacteria on surfaces of trays (one percent by weight).

3.3 Design and dimensions. Trays shall conform to the design and dimensions shown in figures 1 and 2, as applicable.

3.4 Construction. Trays shall be of laminated construction as follows:

(a) Top layer of uncolored glass fiber veil.
(b) Highly reflective gold colored metallized polyester plastic strips and dots (see 3.13.1.2), only when natural color is specified.
(c) Colored glass fiber veil inclusion at manufacturer's option (see 3.2.1.1).
(d) Core with wire reinforced rim (see 3.2.1.2, 3.2.2, 3.4.2, and figure 1, section Y-Y).
(e) Colored glass fiber veil inclusion at manufacturer's option.
(f) Bottom layer of uncolored glass fiber veil.

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The layers shall be bonded together in the order specified under heat and pressure, using the polyester laminating resin specified in 3.2.3. Trays shall not be made with a gel resin overlay nor any other thick resin surface or surfacing material.

3.4.1 *Stacking lugs.* Stacking lugs shall be integrally molded externally on the side wall below the rim of the trays. The lugs shall contact the tray underneath to permit an air space between trays for drying when the trays are stacked. Position of the lugs is shown in figures 1 and 2, as applicable. Minimum lug dimensions shall be 1/4 inch measuring down from the bottom of the rim by 3/8 inch parallel to the rim by 0.040 inch projection.

3.4.2 *Wire ends.* The ends of wire specified in 3.2.2 shall be butt welded in order to make a complete perimeter piece in the tray. Wire shall be molded in position as shown in figure 1, section Y-Y.

3.5 Resistance to boiling water. Trays shall show no cracks, change of color, or blooming of fibers and shall not warp diagonally more than 0.093 inch when tested in accordance with 4.4.4.

3.6 Resistance to impact. The tray shall show no rupture, separated pieces, nor open cracks when tested in accordance with 4.4.1. Tight cracks or crazing shall not be considered rupture.

3.7 Cure. Trays shall show a minimum average dial reading of 30 when tested with the Barcol Impressor in accordance with 4.4.2.

3.8 Warp. Warp shall be not greater than 0.0625 inch when measured in accordance with 4.4.3 and 4.4.3.1.

3.9 Resistance to dry heat. The trays shall show no cracks nor blisters when tested in accordance with 4.4.5.

3.10 Odor. Trays as received shall have no objectionable odor when tested in accordance with 4.4.6.

3.11 Resistance to acids. Trays shall show no change in finish after being tested in accordance with 4.4.7.

3.12 Resistance to alkaline solutions. Trays shall show no signs of cracking nor change in surface appearance after being tested in accordance with 4.4.8.

3.13 Color.

3.13.1 *Type I.* The color for type I trays shall be either tan or natural color as specified (see 6.2).

3.13.1.1 *Tan color.* Tan color shall conform to one of the color chip numbers 20313, 20318, 20372, or 20450 of Fed. Std. No. 595 when tested as specified in 4.4.9.

3.13.1.2 *Natural color.* Natural color shall conform to either color chip 27778 or 27855 of Fed. Std. No. 595 when tested as specified in 4.4.9. In addition, prior to placement of the uncolored top glass fiber veil, approximately one tablespoon of strips, and approximately one teaspoon of specks or dots made from a highly reflective gold colored metallized polyester plastic, shall be scattered at random over the uppermost exposed glass fibers. When the tray is completed, the gold colored metallized polyester plastic shall show through the uncolored top glass fiber veil. The strips shall be approximately 1-1/2 inches long by 0.020 inch wide by 0.0008 inch thick; the specks or dots shall be approximately 0.025 inch in diameter by 0.0008 inch thick. The gold color shall conform to color chip number 17043 of Fed. Std. No. 595 when tested in accordance with 4.4.9.

3.13.2 *Type II.* The colors for this tray shall be blue, green, ivory, or red in accordance with samples at the procurement office (see 6.2).

3.14 Surface finish. The surface of the tray shall have a durable smoothness characteristic of the molded finish and not formed by application of any additional process or coating.

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3.15 **Gloss.** The surface of the tray shall have a specular gloss value of 55 minimum to 70 maximum when tested in accordance with 4.4.9.1.

3.16 **Marking.** Trays shall be permanently and distinctly marked on the outside bottom surface with the manufacturer's trademark or brand name. The marking shall be incorporated in such a manner that the surface opposite the marking shall show no signs of surface penetration, surface roughness, or cause distortion of any part of the tray.

3.17 **Workmanship.** The finished tray shall have smoothly rounded edges and shall be uniform in texture. Trays shall be clean and free from cracks, breaks, abraded areas, and areas of delamination or any other defect which will affect the serviceability for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Preproduction sample inspection.* When preproduction samples are specified (see 6.2), they shall be inspected and tested to determine conformance with all the requirements of this specification.

4.1.2 *Inprocess inspection.* Each lot shall be inspected for conformance with the lamination requirements in 3.4.

4.2 **Sampling.**

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of all trays of the same type and size submitted for inspection at the same time.

4.2.2 *Dimensional and visual examination.* Sampling for these characteristics shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 *Tests.* Sampling for tests shall be in accordance with level S-2 in MIL-STD-105.

4.3 **Inspection.**

4.3.1 *End item.* Trays selected in 4.2.2 shall be examined in accordance with table I. The acceptable quality level (AQL) shall be 2.5 percent for major defects and 6.5 percent for total defects, expressed in percent defective, in accordance with MIL-STD-105.

TABLE I. *Classification of defects, end item*

Defects	Major	Minor
Scuffed or abraded	X	
Scratches	X	
Dents		X
Cracks or fractures	X	
Pinholes	X	
Blisters	X	
Wrinkles		X
Pits	X	
Flow marks		X
Shrinkage	X	
Unfilled area	X	
Chipped	X	
Presence of mechanical oil or grease, dirt, wood shavings, metal chips, or other foreign matter		X
Any sign of delamination	X	
Lacks luster (dull)	X	
Not uniform high luster throughout		X
Not specified color	X	
Lack of uniformity in texture and finish		X
Presence of any finishing coating	X	
Any objectionable odor	X	
Varies from design and shape shown on applicable drawing	X	
Corners and edges not broken and polished		X
Like units fail to stack	X	
Poor nesting resultive in restricted air circulation (for drying purposes)	X	

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TABLE I. *Classification of defects, end item (cont'd)*

Defects	Major	Minor
Illegibly marked or incorrectly marked	X	
Marking not on outside bottom surface	X	
Any dimension not as specified	X	
Damage or defect not affecting function or serviceability		X
Damage or defect affecting function or serviceability	X	

4.3.2 *Inspection of preparation for delivery requirements.* An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE II. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.3.3 *Testing of end item.* The AQL, in accordance with MIL-STD-105, for testing samples selected in 4.2.3, shall be 4.0 percent defective except as otherwise stated herein.

4.4 **Tests.**

4.4.1 *Impact resistance.* In position of normal use, trays shall be supported at four corners (for type II trays see 4.4.1.1) by wooden blocks positioned with faces 2-1/2 inches from the tray edges. The blocks shall be 1 inch square and 3 inches high with the tray bearing ends rounded to approximately 1/2 inch radius. An iron or steel ball weighing 8 pounds (see 6.4), shall be dropped from a height of 18 inches, measured from the bottom of the ball to the tray, onto the center of the tray. If the ball bounces it must be caught by hand so that it strikes the tray only once. This drop shall be repeated three times giving a total of four drops. The tray shall be examined after each drop for any signs of rupture, separated pieces, or open cracks.

4.4.1.1 *Type II support.* Type II tray support for the impact test shall be at the two outer corners which are 18 inches apart and the two outer corners which are 8 inches apart.

4.4.2 *Cure.* Measurements for degree of cure shall be made using the Barcol Impressor. Five readings shall be taken at each of the corners and at the center of the tray on both the inside and bottom surfaces. A metal block placed beneath the surface being tested should give the necessary support needed for accurate readings. Results for each set of five readings shall be averaged.

4.4.3 *Warp.* Rim warp shall be determined by placing the tray upside down on a flat surface and holding it firmly in place by a 10 pound weight with a diameter not exceeding 5 inches positioned at the center of the tray. In this position, the tray shall be sufficiently plane so that the clearance between the tray and the flat surface is less than the thickness of a 0.0625 inch maximum feeler gage.

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4.4.3.1 *Warp (other than rim).* Trays shall be placed upright (position of normal use on the table) on a flat surface and loaded with a 10 pound weight as specified in 4.4.3. In this position, the tray shall be sufficiently plane so that the clearance between the tray and the flat surface is less than the thickness of a 0.0625 inch maximum feeler gage.

4.4.4 *Resistance to boiling water.* Each sample tray shall be immersed by suspension for 30 minutes in boiling water, removed, and allowed to stand for one hour at room temperature. This cycle shall be immediately repeated three times for a total of four such cycles. The sample shall then be allowed to stand overnight in air at normal temperature before examination. Diagonal warp shall be determined by placing the tray (position of normal use on table), on a flat surface. If diagonal corners are found not touching the flat surface, either raised corner shall be held down, by hand, so that three corners are touching. In this position, the clearance between the raised corner of the tray and the flat surface shall be less than the thickness of a 0.093 inch maximum feeler gage.

4.4.5 *Resistance to dry heat.* Place the tray in a circulating air oven for 8 hours at 170 ± 3°F. The specimen shall then be removed, placed in the position of use on a table, and allowed to cool normally to room temperature before examination for cracks or blisters for conformance with 3.9.

4.4.6 *Odor.* The trays shall be half filled with boiling distilled water. A sheet of plate glass shall be immediately placed over the rim of the tray and weighted down to obtain good contact between rim and glass. After 5 minutes, the sheet of glass shall be removed and the presence or absence of odor shall be determined after waiting 60 ± 3 seconds.

4.4.7 *Resistance to acids.* Place one pool of 5 percent acetic acid, one of 5 percent citric acid, and one of 5 percent hydrochloric acid (all by weight), on the top and bottom surfaces of three trays. Each pool, consisting of approximately 15 drops of solution, spaced sufficiently apart to prevent mixing of the solutions. Pools of solution to remain on the trays for 4 hours and then removed and the surface wiped dry. Any change in surface finish indicating a softening, cracking, loss of material, or color in one tray is cause for rejection of the entire delivered lot.

4.4.8 *Resistance to alkaline solutions.* Immerse three trays in a 2 percent sodium carbonate solution and boil for 30 minutes. Remove trays, wipe dry, and examine for signs of cracking or other changes to the surface that affects appearance and serviceability. Any such change on one tray shall be cause for rejection of the entire lot.

4.4.9 *Color.* Colors of trays examined shall conform to either the samples or color chips indicated in 3.1, 3.13.1.1, 3.13.1.2, or 3.13.2, as specified in 6.2, when samples are placed on the tray and compared at arms length in daylight shaded from sunlight.

4.4.9.1 *Gloss.* Gloss determination shall be made in accordance with method 6101 of Fed. Std. No. 141, and results shall conform to the requirements of 3.15.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Twelve trays of like description shall be nested with a layer of paper conforming to UU-P-553, type I, inserted between each tray, extending over the outer edges of the tray. The stack of trays shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 *Level B.* Twelve trays of like description shall be nested with a layer of paper conforming to UU-P-553, type I, inserted between each tray, extending over the

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outer edges of the tray. The stack of trays shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 *Level C.* The trays shall be packaged in accordance with the supplier's commercal practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Unit packages of trays of like description in quantities as specified (see 6.2), shall be packed in a close-fitting box conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant, or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of the triple-wall box or the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specificaton.

5.2.2 *Level B.* Four unit packages of trays of like description shall be packed in a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 *Level C.* The trays shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 **Standard packaging (civil agencies).** The standard packaging and packing for civil agencies shall be as specified in 5.1.2, and 5.2.2, respectively.

5.4 **Marking.**

5.4.1 *Civil agencies.* In addition to markings required by the contract or order, the interior package and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military activities.* In addition to markings required by the contract or order, The interior package and shipping containers shall be marked in accordance with MIL-STD-129.

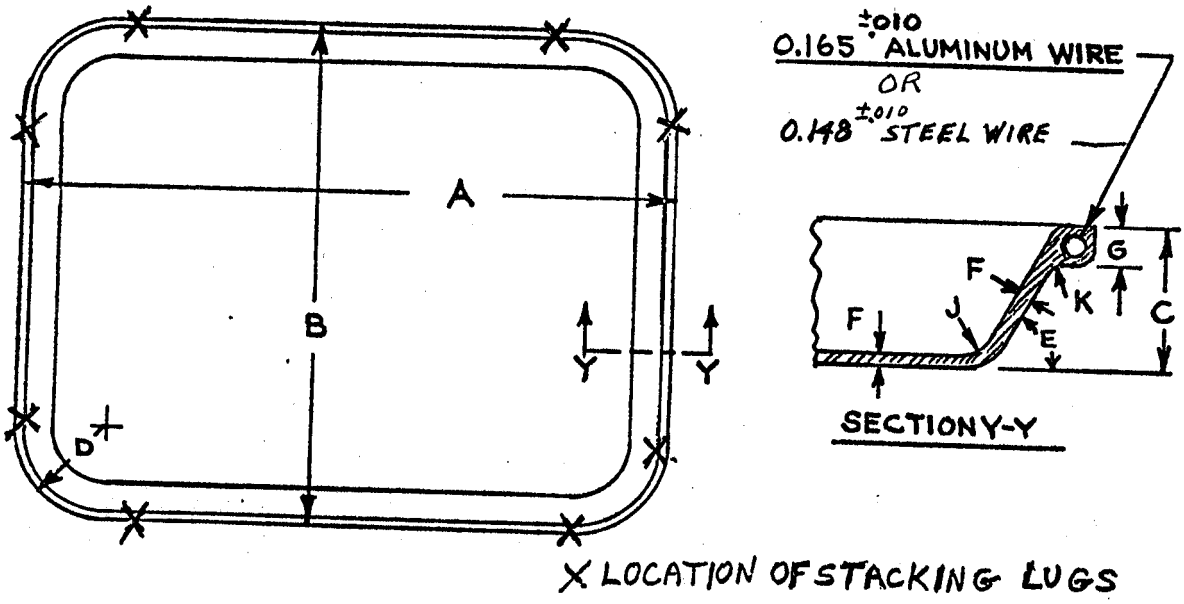
6. NOTES

6.1 **Intended use.** The trays are intended for use as a hand held carrier for dishes containing food in cafeterias, mess halls, and hospitals.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and size (see 1.2).
(*c*) Whether preproduction sample is required (see 3.1).
(*d*) Color required (see 3.13.1, 3.13.2 and 6.3).
(*e*) Size of lot, if different from 4.2.1.
(*f*) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(*g*) Quantities of unit packages required in shipping container (see 5.2.1).

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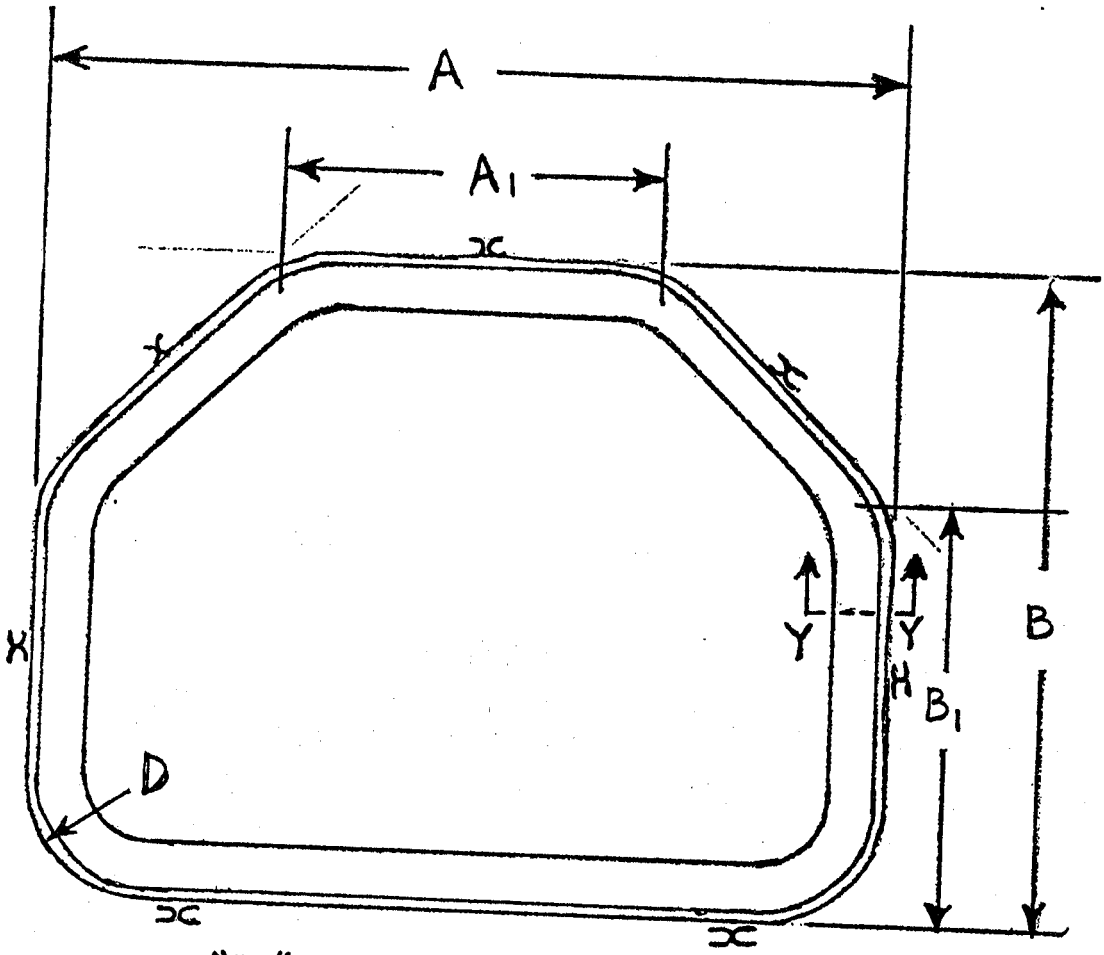


DIMENSIONS OF TRAY, SERVICE (FIBER GLASS), RECTANGULAR TYPE I

FIGURE 1.

Size	A Length ± 3/32 inches	B Width ± 3/32 inches	C Depth ± 3/32 inch	D Corner radius ± 1/16 inches	E Side wall angle ± 2 degrees	F Side wall and bottom thickness ± 0.012 inch	G Lip thick- ness ± 0.020 inch	J Radius at base ± 1/32 inch	K Radius at lip ± 1/32 inch
1	20-1/2	15-1/4	7/8	2-9/16	55	0.093	0.250	5/16	1/8
2	18	14-1/16	7/8	2-9/16	55	.093	.250	5/16	1/8

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TYPE II TRAY, SERVICE (FIBER GLASS), TRAPEZOID

FIGURE 2.

Size	A Length ± 3/32 inches	B Width ± 3/32 inches	C Depth ± 3/32 inch	D Corner radius ± 1/16 inches	E Side wall angle ± 2 degrees	F Side wall and bottom thickness ± 0.012 inch	G Lip thick- ness ± 0.20 inch	J Radius at base ± 1/32 inch	K Radius at lip ± 1/32 inch
1	18	14	7/8	2-5/8	55	0.093	0.250	5/16	1/8
	A1 Length ± 3/32 Inch	B1 Length ± 3/32 Inch							
	8	9							

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6.3 A set of standard color control samples for type II trays shall be available at the procuring agency for examination by bidders.

6.4 An 8 pound iron or steel ball specified in 4.4.2, can be readily obtained from distributors of athletic goods (standard balls for "putting-the-shot" weigh 8 and 12 pounds).

6.5 Type "E" fiber glass. A lime-aluminum-borosilicate glass that is relatively soda free. The "E" designation stems from the electrical characteristics of the glass, which also possesses excellent chemical resistance. "E" glass has a virgin monofilament tensile strength of 500,000 to 550,000 p.s.i. and usable strength of approximately 400,000 p.s.i.

Custodians:

Navy—SA
Air Force—67

Review activities:

Navy—SA, MC, MS
Air Force—67

Interested activity:

VA—DMS

Preparing activity:

GSA-FSS

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